

CS 2601 Linear and Convex Optimization

4. Convex functions (part 2)

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Fall 2023

Outline

- Properties of convex functions
- Convexity-preserving operations

Global minima of convex functions

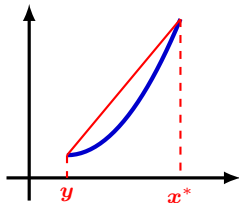
Theorem. Let f be a convex function defined over a convex set S . If $\mathbf{x}^* \in S$ is a local minimum of f , then it is also a global minimum of f over S .

Proof. Suppose there exists $\mathbf{y} \in S$ and $\mathbf{y} \neq \mathbf{x}^*$ s.t. $f(\mathbf{y}) < f(\mathbf{x}^*)$. For $\theta \in (0, 1)$, let $\mathbf{x}_\theta = \theta\mathbf{y} + \bar{\theta}\mathbf{x}^*$. Then

$$f(\mathbf{x}_\theta) \leq \theta f(\mathbf{y}) + \bar{\theta} f(\mathbf{x}^*) < \theta f(\mathbf{x}^*) + \bar{\theta} f(\mathbf{x}^*) = f(\mathbf{x}^*)$$

But $\mathbf{x}_\theta \in S$ by convexity of S , and

$$\|\mathbf{x}_\theta - \mathbf{x}^*\| = \theta \|\mathbf{x}^* - \mathbf{y}\| \rightarrow 0 \quad \text{as} \quad \theta \rightarrow 0$$



contradicting the assumption that $\mathbf{x}^*$ is a local minimum.

Note. This theorem does **not** assert the existence of a global minimum in general! It assumes the existence of a local minimum to start with.

Example. $f(x) = e^x$ has no global or local minimum over $\mathbb{R}$.

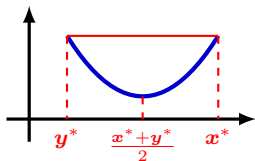
Global minima of convex functions (cont'd)

Theorem. Let f be a strictly convex function defined over a convex set S . If $\mathbf{x}^* \in S$ is a global minimum of f , then it is unique.

Proof. Suppose there exists $\mathbf{y}^* \in S$ and $\mathbf{y}^* \neq \mathbf{x}^*$ s.t. $f(\mathbf{y}^*) = f(\mathbf{x}^*)$. By strict convexity,

$$f\left(\frac{\mathbf{x}^* + \mathbf{y}^*}{2}\right) < \frac{1}{2}f(\mathbf{x}^*) + \frac{1}{2}f(\mathbf{y}^*) = f(\mathbf{x}^*)$$

contradicting the global optimality of $\mathbf{x}^*$.



Note. Strict convexity is a sufficient condition for unique global minimum, but it is **not** necessary!

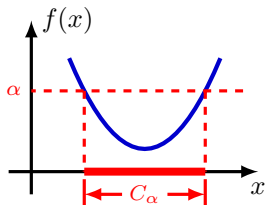
Example. $f(x) = |x|$ has a unique global minimum $x^* = 0$, but it is not strictly convex.

Note. Similar results hold for maxima of concave functions.

Sublevel sets

The α -sublevel set of a function f is

$$C_\alpha = \{\mathbf{x} \in \text{dom} f : f(\mathbf{x}) \leq \alpha\}$$



Theorem. Sublevel sets of a convex function are convex.

Note. For concave f , superlevel set $\{\mathbf{x} \in \text{dom} f : f(\mathbf{x}) \geq \alpha\}$ is convex.

Examples.

- Halfspace $H = \{\mathbf{x} : \mathbf{w}^T \mathbf{x} \leq b\}$
- Norm ball $\bar{B}(\mathbf{x}_0, r) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| \leq r\}$
- Ellipsoid $\mathcal{E} = \{\mathbf{x}_0 + \mathbf{A}\mathbf{u} : \|\mathbf{u}\|_2 \leq 1\}$, $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{A} \succ \mathbf{O}$.

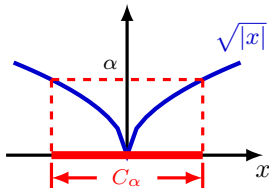
$$\mathcal{E} = \{\mathbf{x} : f(\mathbf{x}) \leq 1\}, \quad f(\mathbf{x}) = \|\mathbf{A}^{-1}(\mathbf{x} - \mathbf{x}_0)\|_2^2 = (\mathbf{x} - \mathbf{x}_0)^T \mathbf{A}^{-2} (\mathbf{x} - \mathbf{x}_0)$$

Sublevel sets (cont'd)

The converse is **not** true. Nonconvex functions can have convex sublevel sets.

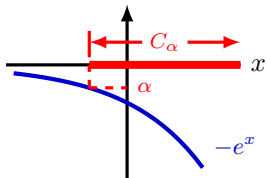
Example. $f(x) = \sqrt{|x|}$ is not convex, but its sublevel sets are all convex,

$$C_\alpha = \begin{cases} \emptyset, & \text{if } \alpha < 0 \\ [-\alpha^2, \alpha^2], & \text{if } \alpha \geq 0 \end{cases}$$



Example. $f(x) = -e^x$ is strictly concave. Its sublevel sets are all convex,

$$C_\alpha = \begin{cases} \emptyset, & \text{if } \alpha \geq 0 \\ [\log(-\alpha), \infty), & \text{if } \alpha < 0 \end{cases}$$



Question. Is the **level set** $\{\mathbf{x} \in \text{dom} f : f(\mathbf{x}) = \alpha\}$ convex?

Epigraph

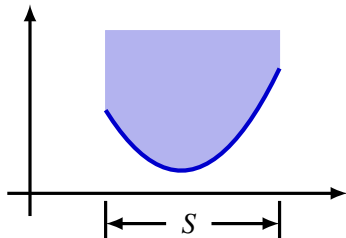
Recall the graph of $f : S \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is the set

$$\{(\mathbf{x}, f(\mathbf{x})) \in \mathbb{R}^{n+1} : \mathbf{x} \in S\}$$

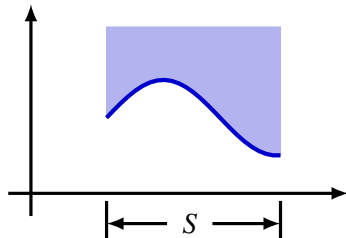
The **epigraph**¹ of f is

$$\text{epi} f = \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} : \mathbf{x} \in S, y \geq f(\mathbf{x})\}$$

Note. f and its extended-value extension $\tilde{f}$ have the same epigraph.



convex



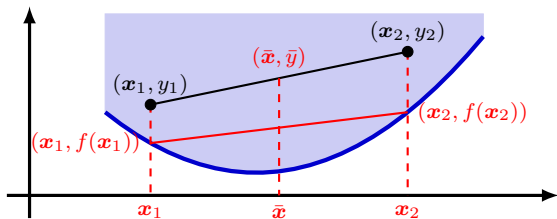
nonconvex

¹The prefix **epi-** means “above”, “over”.

Epigraph (cont'd)

Theorem. $f : S \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function iff $\text{epi} f$ is a convex set.

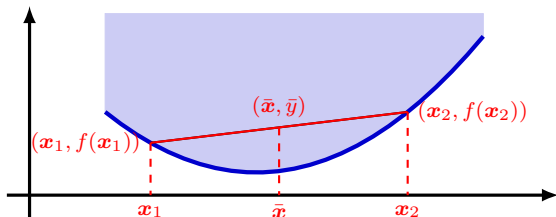
Proof. “ $\Rightarrow$ ”. Assume f is convex. Let $(x_1, y_1), (x_2, y_2) \in \text{epi} f$, $\theta \in [0, 1]$.
Need to show $(\bar{x}, \bar{y}) \triangleq (\theta x_1 + \bar{\theta} x_2, \theta y_1 + \bar{\theta} y_2) \in \text{epi} f$.



1. f convex $\Rightarrow \bar{x} \in S$ and $f(\bar{x}) \leq \theta f(x_1) + \bar{\theta} f(x_2)$
2. $(x_i, y_i) \in \text{epi} f \Rightarrow f(x_i) \leq y_i \Rightarrow \theta f(x_1) + \bar{\theta} f(x_2) \leq \theta y_1 + \bar{\theta} y_2 = \bar{y}$
3. By 1 and 2, $\bar{x} \in S$ and $f(\bar{x}) \leq \bar{y} \Rightarrow (\bar{x}, \bar{y}) \in \text{epi} f$

Epigraph (cont'd)

Proof (cont'd). “ $\Leftarrow$ ”. Assume $\text{epi}f$ is convex. Let $x_1, x_2 \in S$, $\theta \in [0, 1]$. Need to show $\bar{x} \triangleq \theta x_1 + \bar{\theta} x_2 \in S$ and $f(\bar{x}) \leq \theta f(x_1) + \bar{\theta} f(x_2) \triangleq \bar{y}$.



1. $f(x_i) \leq f(x_i) \implies (x_i, f(x_i)) \in \text{epi}f$ by definition
2. $\text{epi}f$ convex $\implies (\bar{x}, \bar{y}) = \theta(x_1, f(x_1)) + \bar{\theta}(x_2, f(x_2)) \in \text{epi}f$
3. $\bar{x} \in S$, $f(\bar{x}) \leq \bar{y} = \theta f(x_1) + \bar{\theta} f(x_2)$ by definition of $\text{epi}f$

Note. The same proof shows the following result: the projection $\{x : (x, y) \in C \text{ for some } y\}$ of a convex set C is convex.

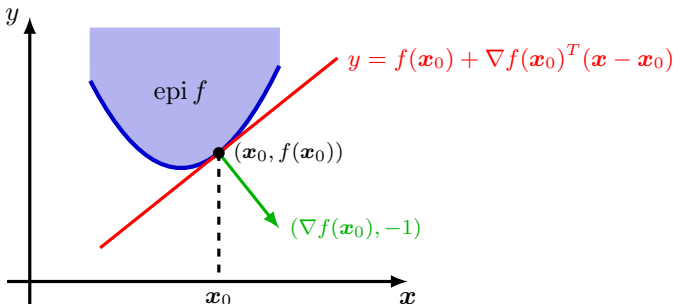
Epigraph (cont'd)

The first-order condition

$$f(\mathbf{x}) \geq f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0)$$

implies $y = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0)$ is a supporting hyperplane of $\text{epi} f$ at $(\mathbf{x}_0, f(\mathbf{x}_0))$, i.e.

$$\begin{bmatrix} \nabla f(\mathbf{x}_0) \\ -1 \end{bmatrix}^T \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} \leq \begin{bmatrix} \nabla f(\mathbf{x}_0) \\ -1 \end{bmatrix}^T \begin{bmatrix} \mathbf{x}_0 \\ f(\mathbf{x}_0) \end{bmatrix}, \quad \forall (\mathbf{x}, y) \in \text{epi} f$$



Jensen's inequality

For convex function f , $\mathbf{x}, \mathbf{y} \in \text{dom} f$, $\theta \in [0, 1]$

$$f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$$

More generally, for $\mathbf{x}_i \in \text{dom} f$, $\theta_i \geq 0$, and $\sum_{i=1}^m \theta_i = 1$,

$$f\left(\sum_{i=1}^m \theta_i \mathbf{x}_i\right) \leq \sum_{i=1}^m \theta_i f(\mathbf{x}_i)$$

Example. $f(x) = x^2$ is convex over $\mathbb{R}$.

$$\left(\sum_{i=1}^n \frac{1}{n} x_i\right)^2 \leq \sum_{i=1}^n \frac{1}{n} x_i^2 \implies \frac{1}{n} \sum_{i=1}^n x_i \leq \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}$$

Example. $f(x) = \log x$ is concave over $(0, \infty)$. For $x_i > 0$,

$$\log\left(\sum_{i=1}^n \frac{1}{n} x_i\right) \geq \sum_{i=1}^n \frac{1}{n} \log x_i \implies \frac{1}{n} \sum_{i=1}^n x_i \geq \sqrt[n]{\prod_{i=1}^n x_i}$$

Hölder's inequality

Let $p, q \in (1, \infty)$ be **conjugate exponents**, i.e. $p^{-1} + q^{-1} = 1$. For $\mathbf{x} = (x_1, \dots, x_n)^T$, $\mathbf{y} = (y_1, \dots, y_n)^T$, **Hölder's inequality** holds,

$$\sum_{i=1}^n |x_i y_i| \leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q$$

Proof. Assume $\mathbf{x} \neq \mathbf{0}, \mathbf{y} \neq \mathbf{0}$; otherwise trivial. Let $\tilde{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|_p$ and $\tilde{\mathbf{y}} = \mathbf{y}/\|\mathbf{y}\|_q$. The above inequality is equivalent to $\sum_{i=1}^n |\tilde{x}_i \tilde{y}_i| \leq 1$.

1. Show $x^{\frac{1}{p}} y^{\frac{1}{q}} \leq \frac{1}{p}x + \frac{1}{q}y$ for $x, y \geq 0$.

▶ trivial if $xy = 0$

▶ if $xy > 0$, $\log x$ is concave $\implies \log\left(\frac{1}{p}x + \frac{1}{q}y\right) \geq \frac{1}{p}\log x + \frac{1}{q}\log y$

2. Let $x = |\tilde{x}_i|^p$ and $y = |\tilde{y}_i|^q$ in the inequality in 1,

$$|\tilde{x}_i| \cdot |\tilde{y}_i| \leq p^{-1}|\tilde{x}_i|^p + q^{-1}|\tilde{y}_i|^q$$

3. Sum over i and note $\|\tilde{\mathbf{x}}\|_p = \|\tilde{\mathbf{y}}\|_q = 1$,

$$\sum_{i=1}^n |\tilde{x}_i \tilde{y}_i| \leq \frac{1}{p} \|\tilde{\mathbf{x}}\|_p^p + \frac{1}{q} \|\tilde{\mathbf{y}}\|_q^q = \frac{1}{p} + \frac{1}{q} = 1$$

Minkowski's inequality

For $1 < p < \infty$,

$$\|\mathbf{x} + \mathbf{y}\|_p \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$$

Proof. Only need to consider case $\|\mathbf{x} + \mathbf{y}\|_p > 0$.

- $\|\mathbf{x} + \mathbf{y}\|_p^p = \sum_i |x_i + y_i|^p \leq \sum_i |x_i| \cdot |x_i + y_i|^{p-1} + \sum_i |y_i| \cdot |x_i + y_i|^{p-1}$
- Let $\frac{1}{p} + \frac{1}{q} = 1$. Applying Hölder's inequality and $(p-1)q = p$,

$$\sum_i |x_i| \cdot |x_i + y_i|^{p-1} \leq \|\mathbf{x}\|_p \left(\sum_i |x_i + y_i|^{(p-1)q} \right)^{1/q} = \|\mathbf{x}\|_p \|\mathbf{x} + \mathbf{y}\|_p^{p/q}$$

- Interchanging x and y , $\sum_i |y_i| \cdot |x_i + y_i|^{p-1} \leq \|\mathbf{y}\|_p \|\mathbf{x} + \mathbf{y}\|_p^{p/q}$
- Combining above inequalities,

$$\|\mathbf{x} + \mathbf{y}\|_p^p \leq (\|\mathbf{x}\|_p + \|\mathbf{y}\|_p) \|\mathbf{x} + \mathbf{y}\|_p^{p/q}$$

- Cancel $\|\mathbf{x} + \mathbf{y}\|_p^{p/q}$ and note $p - p/q = 1$.

Outline

- Properties of convex functions
- Convexity-preserving operations

Convexity-preserving operations

- nonnegative combinations

$$f(\mathbf{x}) = \sum_{i=1}^m c_i f_i(\mathbf{x})$$

- composition with affine functions

$$f(\mathbf{x}) = g(\mathbf{Ax} + \mathbf{b})$$

- composition of convex/concave functions

$$f(\mathbf{x}) = h(g_1(\mathbf{x}), \dots, g_m(\mathbf{x}))$$

- pointwise maximum/supremum

$$f(\mathbf{x}) = \sup_{i \in I} f_i(\mathbf{x})$$

- partial minimization

$$f(\mathbf{x}) = \inf_{\mathbf{y} \in C} g(\mathbf{x}, \mathbf{y})$$

Nonnegative combinations

Proposition. Let $f_i : \mathbb{R}^n \rightarrow (-\infty, \infty]$, $i = 1, \dots, m$, be convex functions. Then for any $c_1, \dots, c_m \geq 0$, the function $f : \mathbb{R}^n \rightarrow (-\infty, \infty]$ defined by

$$f(\mathbf{x}) = \sum_{i=1}^m c_i f_i(\mathbf{x})$$

is convex, with $\text{dom} f = \bigcap_{i=1}^m \text{dom} f_i$ ².

Questions. Let f_1, f_2 be convex functions.

- Is $f_1 - f_2$ convex?
- Is $f_1 \cdot f_2$ convex?
- Is $\frac{f_1}{f_2}$ convex?

²Often we require $\text{dom} f \neq \emptyset$ to preclude the trivial case $f(\mathbf{x}) = \infty, \forall \mathbf{x}$

Affine composition

Proposition. Let $g : \mathbb{R}^n \rightarrow (-\infty, \infty]$ be convex, $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{b} \in \mathbb{R}^n$. Then $f : \mathbb{R}^m \rightarrow (-\infty, \infty]$ defined by

$$f(\mathbf{x}) = g(\mathbf{Ax} + \mathbf{b})$$

is convex, with $\text{dom } f = \{\mathbf{x} \in \mathbb{R}^m : \mathbf{Ax} + \mathbf{b} \in \text{dom } g\}$.

Example. $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{y}\|$ is convex

Example. The following log-sum-exp function is convex

$$f(\mathbf{x}) = \log \left(\sum_{i=1}^n e^{\mathbf{w}_i^T \mathbf{x} + b_i} \right).$$

Take $g(\mathbf{y}) = \log(\sum_{i=1}^n e^{y_i})$ and $\mathbf{y} = [\mathbf{w}_1, \dots, \mathbf{w}_n]^T \mathbf{x} + \mathbf{b}$.

Example. $f(x_1, x_2) = (x_1 - 2x_2)^4 + 2e^{3x_1 + 2x_2 - 5}$ is convex.

Take $g(y_1, y_2) = y_1^4 + 2e^{y_2}$ and $(y_1, y_2) = (x_1 - 2x_2, 3x_1 + 2x_2 - 5)$.

Scalar composition

Proposition. Consider $g : \mathbb{R}^n \rightarrow \mathbb{R}$, $h : \mathbb{R} \rightarrow \mathbb{R}$ and $f(\mathbf{x}) = h(g(\mathbf{x}))$.

- f is convex if h is convex and increasing, g is convex
- f is convex if h is convex and decreasing, g is concave
- f is concave if h is concave and increasing, g is concave
- f is concave if h is concave and decreasing, g is convex

Note. When h is increasing, f , h , g have the same convexity property. When h is decreasing, f , h , $-g$ have the same convexity property. For $n = 1$ and differentiable g and h , this can be seen from

$$f''(x) = h''(g(x))[g'(x)]^2 + h'(g(x))g''(x)$$

Proof of the first case. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\theta \in [0, 1]$.

1. g convex $\implies g(\theta\mathbf{x} + \bar{\theta}\mathbf{y}) \leq \theta g(\mathbf{x}) + \bar{\theta}g(\mathbf{y})$
2. h increasing $\implies h \circ g(\theta\mathbf{x} + \bar{\theta}\mathbf{y}) \leq h(\theta g(\mathbf{x}) + \bar{\theta}g(\mathbf{y}))$
3. h convex $\implies h(\theta g(\mathbf{x}) + \bar{\theta}g(\mathbf{y})) \leq \theta h(g(\mathbf{x})) + \bar{\theta}h(g(\mathbf{y}))$
4. 2 and 3 $\implies f(\theta\mathbf{x} + \bar{\theta}\mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta}f(\mathbf{y})$

Scalar composition (cont'd)

Example. $f(\mathbf{x}) = e^{\|\mathbf{x}\|}$ is convex for any norm $\|\cdot\|$. Take $g(\mathbf{x}) = \|\mathbf{x}\|$, $h(x) = e^x$.

Example. $f(\mathbf{x}) = \|\mathbf{x}\|^2$ is convex. Take $g(\mathbf{x}) = \|\mathbf{x}\|$, $h(x) = x^2$ for $x \geq 0$ and $h(x) = 0$ for $x < 0$. (Can we take $h(x) = x^2$ here?)

Example. $f(\mathbf{x}) = e^{\mathbf{x}^T \mathbf{Q} \mathbf{x}}$ is convex if $\mathbf{Q} \succeq \mathbf{0}$. Take $g(\mathbf{x}) = \mathbf{x}^T \mathbf{Q} \mathbf{x}$, $h(x) = e^x$.

Note. When the domains of g and h are not the entire $\mathbb{R}^n$ or $\mathbb{R}$, similar results hold for their extended-value extensions.

Note. When the conditions fail, we need case-by-case analysis.

Example. $g(x) = x^2$ is convex, $h(x) = e^{-x}$ is convex and decreasing, $f(x) = e^{-x^2}$ is neither convex nor concave.

Example. $g(x) = 1 + e^x$ and $h(x) = -\log x$ are convex, the extension $\tilde{h}$ is decreasing, $f(x) = h(g(x)) = -\log(1 + e^x)$ is concave.

Vector composition

$h : \mathbb{R}^m \rightarrow \mathbb{R}$ is increasing (decreasing) if

$$\mathbf{x} \geq \mathbf{y} \text{ (componentwise)} \implies h(\mathbf{x}) \geq (\leq) h(\mathbf{y})$$

Proposition. Let $h : \mathbb{R}^m \rightarrow \mathbb{R}$, $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, 2, \dots, m$, and define $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$f(\mathbf{x}) = h(g_1(\mathbf{x}), \dots, g_m(\mathbf{x}))$$

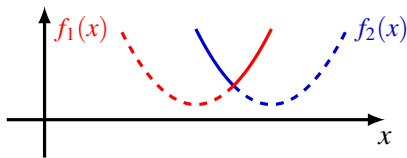
- f is convex if h is convex and increasing, g_i are convex
- f is convex if h is convex and decreasing, g_i are concave
- f is concave if h is concave and increasing, g_i are concave
- f is concave if h is concave and decreasing, g_i are convex

Pointwise maximum

Proposition. If $f_i : \mathbb{R}^n \rightarrow (-\infty, \infty]$ be convex functions, $i = 1, 2, \dots, m$. Then the pointwise maximum $f : \mathbb{R}^n \rightarrow (-\infty, \infty]$ defined by

$$f(\mathbf{x}) = \max_{1 \leq i \leq m} f_i(\mathbf{x})$$

is convex, with domain $\text{dom} f = \bigcap_{i=1}^m \text{dom} f_i$.



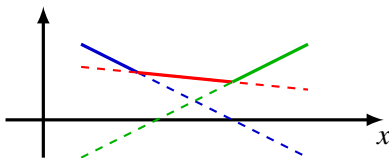
Proof. $\text{dom} f$ is convex. Fix $\mathbf{x}, \mathbf{y} \in \text{dom} f$, $\theta \in [0, 1]$. For each i ,

$$\begin{aligned} f_i(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) &\leq \theta f_i(\mathbf{x}) + \bar{\theta} f_i(\mathbf{y}) && \text{by convexity of } f_i \\ &\leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y}) && \text{by definition of } f, f_i \leq f \end{aligned}$$

Maximizing over i yields $f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$.

Pointwise maximum (cont'd)

Example. $f(\mathbf{x}) = \max_{1 \leq i \leq m} (\mathbf{w}_i^T \mathbf{x} + b_i)$ is convex and piecewise linear.



Special cases.

- Hinge function $(x)^+ = \max\{x, 0\}$

-

$$f(\mathbf{x}) = \max\{x_1, \dots, x_n\} = \max_{1 \leq i \leq n} \mathbf{e}_i^T \mathbf{x},$$

where $\mathbf{e}_i$ is i -th standard basis vector.

Question. Is $f(\mathbf{x}) = \min_{1 \leq i \leq m} f_i(\mathbf{x})$ convex?

Pointwise supremum

Proposition. Let $f_i : \mathbb{R}^n \rightarrow (-\infty, \infty]$, $i \in I$, be convex functions. Then $f : \mathbb{R}^n \rightarrow (-\infty, \infty]$ defined by

$$f(\mathbf{x}) = \sup_{i \in I} f_i(\mathbf{x})$$

is convex.

Proof. Recall the intersection of convex sets is convex.

$$\text{epi } f = \left\{ (\mathbf{x}, y) : y \geq f(\mathbf{x}) = \sup_{i \in I} f_i(\mathbf{x}) \right\} = \bigcap_{i \in I} \{ (\mathbf{x}, y) : y \geq f_i(\mathbf{x}) \} = \bigcap_{i \in I} \text{epi } f_i$$

Example. Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$\phi(\boldsymbol{\lambda}) = \sup_{\mathbf{x} \in \mathbb{R}^n} \{ f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) \}$$

is convex, since $\phi_{\mathbf{x}}(\boldsymbol{\lambda}) \triangleq f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x})$ is affine in $\boldsymbol{\lambda}$ for fixed $\mathbf{x} \in \mathbb{R}^n$.

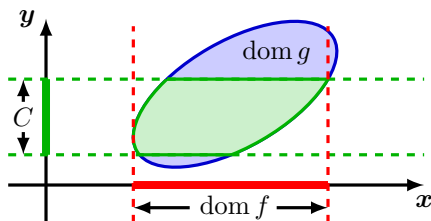
Partial minimization

Proposition. If $g : \mathbb{R}^n \times \mathbb{R}^m \rightarrow (-\infty, \infty]$ is convex, and $\emptyset \neq C \subset \mathbb{R}^m$ is convex, then f defined below is convex provided $f(x) > -\infty$ for all x ,

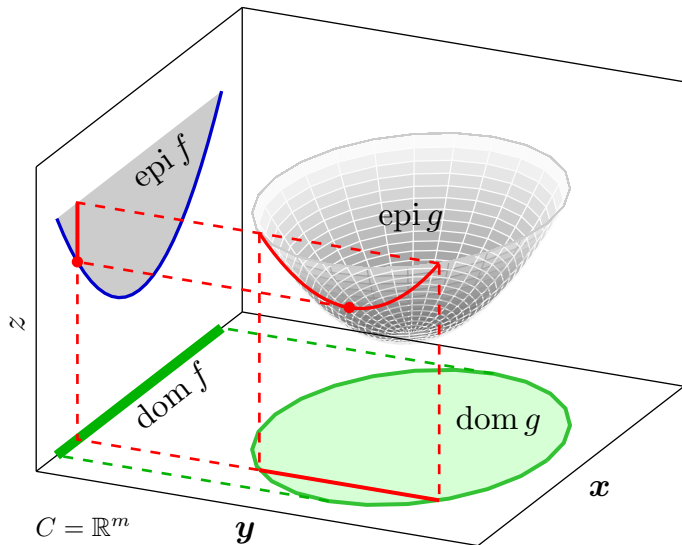
$$f(x) = \inf_{y \in C} g(x, y)$$

Note $\text{dom } f$ is the projection of $\text{dom } g \cap (\mathbb{R}^n \times C)$ onto the x coordinates.

$$\begin{aligned}\text{dom } f &= \{x \in \mathbb{R}^n : g(x, y) < \infty \text{ for some } y \in C\} \\ &= \{x \in \mathbb{R}^n : (x, y) \in \text{dom } g \text{ for some } y \in C\}\end{aligned}$$



Partial minimization (cont'd)



Partial Minimization (cont'd)

Proof of proposition. Let $\mathbf{x}_1, \mathbf{x}_2 \in \text{dom } f$ and $\theta \in [0, 1]$.

1. By definition of f , for any $\epsilon > 0$, there exists $\mathbf{y}_i \in C$ s.t.

$$g(\mathbf{x}_i, \mathbf{y}_i) < f(\mathbf{x}_i) + \epsilon, \quad i = 1, 2$$

2. By convexity of g and C , $\theta \mathbf{y}_1 + \bar{\theta} \mathbf{y}_2 \in C$, and

$$\begin{aligned} f(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2) &= \inf_{\mathbf{y} \in C} g(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2, \mathbf{y}) \\ &\leq g(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2, \theta \mathbf{y}_1 + \bar{\theta} \mathbf{y}_2) && \theta \mathbf{y}_1 + \bar{\theta} \mathbf{y}_2 \in C \\ &\leq \theta g(\mathbf{x}_1, \mathbf{y}_1) + \bar{\theta} g(\mathbf{x}_2, \mathbf{y}_2) && g \text{ is convex} \\ &< \theta [f(\mathbf{x}_1) + \epsilon] + \bar{\theta} [f(\mathbf{x}_2) + \epsilon] && \text{by step 1} \\ &= \theta f(\mathbf{x}_1) + \bar{\theta} f(\mathbf{x}_2) + \epsilon \end{aligned}$$

3. Since ϵ is arbitrary,

$$f(\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2) \leq \theta f(\mathbf{x}_1) + \bar{\theta} f(\mathbf{x}_2)$$

so f is convex

Partial minimization (cont'd)

Example. Distance to convex set is convex, $\text{dist}(\mathbf{x}, C) = \inf_{y \in C} \|\mathbf{x} - \mathbf{y}\|$

